

Title: Wildlife Corridor Analysis using Species-Habitat Interaction Matrix

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Introduction:

Habitat fragmentation is a major challenge in wildlife conservation. Animals require safe corridors to move between habitats for survival, food, and breeding. This project focuses on analyzing species-habitat interactions using basic AI/ML techniques.

Objective:

To build a species-habitat interaction matrix and identify similarities between species to understand movement patterns and habitat preferences.

Dataset:

A sample dataset was created containing different animal species and their interaction with habitat zones such as forest, grassland, and water bodies.

Methodology:

- Data was structured in CSV format
 - Interaction matrix was created using Pandas
 - Cosine similarity was applied to measure similarity between species
 - Habitat suggestions were generated based on similar species
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Result:

The system successfully generated an interaction matrix and identified similar species. Based on similarity, new habitat suggestions were provided.

Conclusion:

This project demonstrates how basic machine learning concepts can be applied to ecological problems like wildlife corridor analysis. It provides a foundation for more advanced AI-based conservation systems.